Test 3

Name_____Date_____

Useful equations and Constants	
• $c = \lambda\nu$ • $Moles\ (n) = \frac{Mass}{Molar\ Mass}$ • $E = h\nu$ • $PV = nRT$ • $K = °C + 273$	• $N_A = 6.022 \times 10^{23}\ atoms/mol$ • $h = 6.626 \times 10^{-34}\ J.s$ • $c = 3.0 \times 10^{8}\ \frac{m}{s}$ • 1kJ = 1000 J • 1kg = 1000 g • 1 m = 10^9 nm • $R = 0.08206\ \frac{L.\ atm}{mol\ .K}$

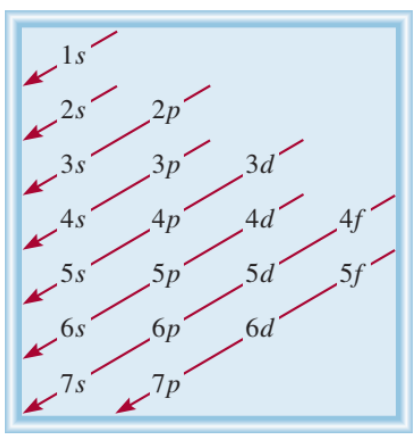


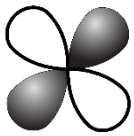
Fig. The order in which atomic subshells are filled in a many-electron atom.

Part I. Choose the most appropriate answer (1 pt. each)

1. According to Planck's hypothesis, the energy of photon is directly proportional to its
   A. Amplitude
   B. Wavelength
   C. Momentum
   D. Frequency
2. Which of the following statements is false?
   A. Principal quantum number n determines the energy of an orbital
   B. Angular quantum number l determines shape of an orbital
   C. Magnetic quantum number m_l determines orientation of an orbital
   D. Spin quantum number m_s determines spin orientation of electrons
   E. None of these
3. Calculate the wavelength of an electron (m = 9.11 × 10-28 g) moving at 3.22 × 106 m/s.
   A. 2.26 × 10-10 m
   B. 4.52 × 10-10 m
   C. 1.81 × 10-10 m
   D. 2.26 × 10-9 m
   E. 2.76 × 10-9 m
4. How many electrons can be described by the quantum numbers $n = 3$, $l = 2$, and $m_l = 0$?
   A. 0
   B. 2
   C. 6
   D. 10
   E. 14
5. Which of the following statements is false?
   A. Energy of electromagnetic radiation increases as its frequency increases
   B. An excited atom can return to its ground state by absorbing electromagnetic radiation
   C. An electron in the $n = 4$ state in the hydrogen atom can go to the $n = 2$ state by emitting electromagnetic radiation at the appropriate frequency

D. Frequency and the wavelength of electromagnetic radiation are inversely proportional to each other

6. In the photoelectric effect, if the intensity of light shone on a metal increase, what will happen?
   A. Ejected electrons will be faster.
   B. There will be more electrons ejected.
   C. Ejected electrons will be slower.
   D. There will be fewer electrons ejected.
7. If $l = 2$, which of the following may be true?
   A. $n = 6$
   B. $m_l = -1$
   C. $m_l = 1$
   D. All of these
8. Which of the following species is represented by the given electron configuration? $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
   A. Calcium
   B. Silicon
   C. Potassium
   D. Strontium
9. What is the correct electronic structure for Mg^{2+}?
   A. $1s^2 2s^2 2p^6 3s^2$
   B. $1s^2 2s^2 2p^5 3s^1$
   C. $1s^2 2s^2 2p^6$
   D. $1s^2 2s^2 2p^4 3s^2$
10. What is the value of l for the orbital shown on the right?



   A. 2
   B. 3
   C. 1
   D. 4
   E. 0
11. True or false: two electrons can simultaneously occupy the same (n, ℓ, mℓ) orbital.

A. True

B. False

12. What is the noble gas core in the electron configuration of Ru?

A. Ne

B. Ar

C. Kr

D. Xe

13. Of the following elements, which has occupied *d* orbitals in its ground-state neutral atoms?

A. Ba

B. Ca

C. Si

D. P

E. Cl

14. For the atoms Li, N, F, and Na, which of the following is the correct order from smallest to largest atomic radius?

A. Na, F, N, Li

B. Na, Li, N, F

C. Li, N, F, Na

D. N, F, Na, Li

E. F, N, Li, Na

15. Which of the following correctly ranks the ionization energies of O, F, Na, S, and Cs from smallest to largest?

A. Cs, Na, S, O, F

B. Cs, S, Na, O, F

C. F, O, Na, S, Cs

D. F, O, S, Na, Cs

E. Na, S, Cs, F, O

16. Which of the following is a correct set of quantum numbers for an electron in a 3*d* orbital?

A. $n = 3$, $l = 0$, $m_l = -1$

B. $n = 3, l = 1, m_l = +3$

C. $n = 3, l = 2, m_l = +3$

D. $n = 3, l = 2, m_l = -2$

17. Rank the following orbitals in order of increasing number of nodes.

A. $1s < 2s < 2p < 3s < 3d$

B. $1s < 2s = 2p < 3s < 3d$

C. $1s < 2s = 2p < 3s = 3d$

D. $1s < 2s < 3s < 2p < 3d$

E. $1s < 2s < 3s < 2p = 3d$

18. Use the data given below to construct a **Born-Haber** cycle to determine the lattice energy of CaO.

	ΔH^0(kJ)
Ca(s) → Ca(g)	193
Ca(g) → Ca^+(g) + e^-	590
Ca^+(g) → $Ca2^+$(g) + e^-	1010
2 O(g) → O2(g)	-498
O(g) + e^- → O^-(g)	-141
O^-(g) + e^- → $O2^-$(g)	878
Ca(s) + 1/2 O2(g) → CaO(s)	-635

A. -3414 kJ

B. +1397 kJ

C. -2667 kJ

D. +3028 kJ

E. -2144 kJ

19. The Lewis structure of phosphoric acid (H_3PO_4) contains

A. 1 double bond, 6 single bonds, and 8 lone pairs.

B. 4 double bonds, and two single bonds.

C. 2 double bonds and 4 single bonds.

D. 6 double bonds, 1 single bond, and 8 lone pairs.

E. 2 double bonds, 5 single bonds, and 4 lone pairs.

20. Use the bond energies provided to estimate $\Delta H°_{rxn}$ for the reaction below.

$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$ $\Delta H°_{rxn}$ =?

Bond	Bond Energy (kJ/mol)
C-C	347
C-H	414
C=C	611
C≡C	837
H-H	436

A. -128 kJ

B. +98 kJ

C. +700 kJ

D. -102 kJ

E. -166 kJ

II. **Show all work.** **Show your work in a neat and legible manner for partial credit. (30 points)**

1. **(2 points)** Calculate the frequency of red light of wavelength 6.50×10^2 nm

2. **(2 points)** Compare the wavelength for an electron (mass = 9.11×10^{-31} kg) traveling at a speed of 1.0×10^7 m/s with that for a ball (mass = 0.10 kg) traveling at 35 m/s.

3. **(5 points)** For principal quantum level $n = 5$, determine the number of allowed subshells (different values of l), and give the designation of each

4. **(4 points)** Write the full ground-state electronic configuration of *Manganese (II)* (Mn^{2+}) ion? Explain why this configuration is particularly stable in transition metal chemistry?

5. **(4 points)** Write the Lewis structure for PCl_5

6. **(5 points)** Give possible Lewis structures for XeO_3, an explosive compound of xenon. Which Lewis structure or structures are most appropriate according to the formal charges?

7. **(8 points)** Write Lewis structure for cyanate ion (OCN^-). How many valid resonance structures can be drawn for the cyanate ion? For each resonance structure, assign formal charges to each atom. Which resonance structure is likely to contribute most to the correct structure of OCN^-. (Hint: use table to summarize your work)